

Wear-It-Virtually -- Experiment Report

Demo Run | 2026-05-01 | Python 3 | OpenCV-based Pipeline

Pipeline Overview

The virtual try-on pipeline runs in 3 stages: **(1) Human Parsing** -- body segmentation, **(2) Pose Estimation** -- 18-keypoint skeleton extraction, **(3) Try-On** -- geometric clothing transfer. In demo mode (no model weights needed) each stage uses OpenCV-based fallbacks: GrabCut segmentation, heuristic keypoints, and Gaussian-blended texture paste.

Stage 1 -- Human Parsing

Demo mode uses **GrabCut** (cv2.grabCut) for foreground body segmentation. The body region is isolated from the background and saved as a binary mask (.mat).



Left: Original person image | Right: GrabCut body segmentation overlay

Stage 2 -- Pose Estimation

Demo mode uses a **heuristic upper-body keypoint detector** (shoulder/chest ellipse + arm points). 18 COCO-format keypoints [x, y, confidence] are saved as .mat and overlaid on the original image.



Skeleton overlaid -- green dots = keypoints, orange lines = limb connections

Stage 3 -- Virtual Try-On (Geometric Blend)

The clothing image is geometrically warped to fit the upper-body region detected in stages 1-2. A **Gaussian-blurred mask** ensures smooth feathering at clothing edges. No neural network inference is required in demo mode.



Left: Target clothing image | Right: Final try-on result

Results Summary

Stage	Input	Output	Method	Size
Human Parsing	example_person.jpg	example_person.mat	GrabCut (OpenCV)	48.2 KB
Pose Estimation	example_person.jpg	example_person.mat	Heuristic keypoints	0.4 KB

Try-On	person + clothing	final.png	Gaussian-blend paste	67.3 KB
Clothing Input	example_clothing.jpg	--	JPEG 18.9 KB	--

Generated automatically by the Wear-It-Virtually demo pipeline (Python 3, OpenCV). Full pipeline with model weights (2 GB+) available -- see README for setup instructions.